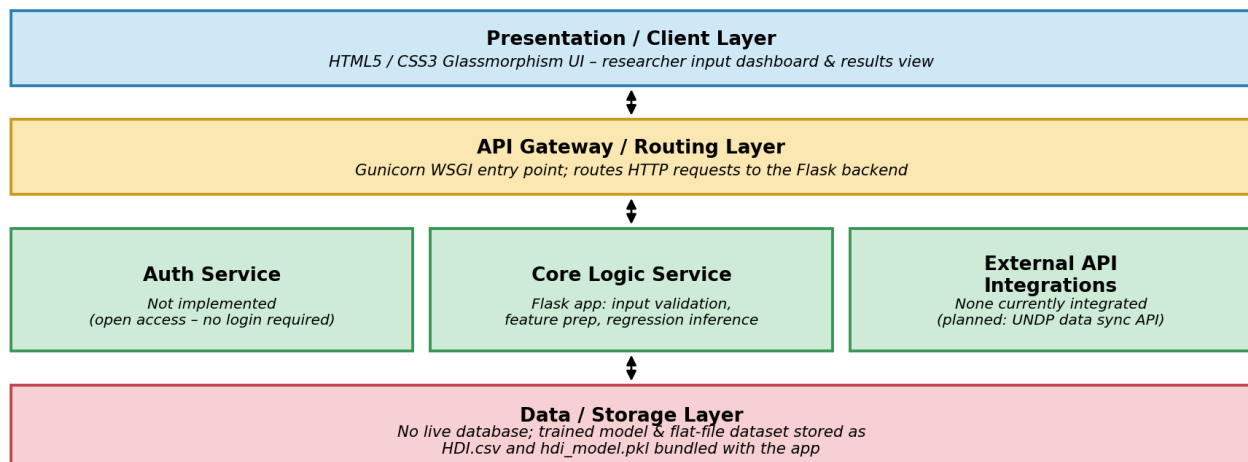


Solution Architecture

Date	1 July 2026
Team ID	SWTID-2026-1614
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

The diagram below reflects the HDI Predictor's deployed architecture: a Flask web application that serves a glassmorphism client UI, queries a local database (HDI.csv) for baseline countries, and runs predictions through a pre-trained scikit-learn regression model.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Renders the glassmorphism dashboard where researchers input Life Expectancy, Schooling, and GNI. Displays predicted HDI score badges, gauges, and hosts the country search autocomplete search box.	HTML5, CSS3, Vanilla JavaScript
API Gateway	Manages web socket routing, receives client HTTP requests, and forwards them to the Python backend logic.	Gunicorn WSGI, Web Hosting router (Render/Vercel)

Component Name	Description / Role in Architecture	Technologies Used
Core Logic Service	Performs incoming form parameter validation, aligns variables into a model-compliant array format, executes Linear Regression model inference, and maps outcomes to development tiers.	Python, Flask, scikit-learn, NumPy, Pandas
Auth Service	Not implemented in the current version. The application is publicly accessible to ensure low-friction access for students and policy analysts.	None (planned: JWT-based admin credentials for dataset uploads)
External API Integrations	No active external APIs called. Country baselines are fetched from the bundled database.	None (planned: UNDP database API integration for automated yearly syncs)
Database / Storage	Holds the historical dataset of 195 countries (HDI.csv) and the serialized Linear Regression model weights (hdi_model.pkl). Both are loaded in-memory at startup for low latency.	Serialized .pkl model file, local HDI.csv flat-file